Angouran Intelligent Exploration Simulator

GPU-Accelerated Synthetic Mineral Exploration AI — Technical Report

What this document is: a factual account of what was built, the measured results from real program runs on real GPU hardware, an assessment of how accurate those results are, and an honest answer to “can we do this for real at Angouran.” No number in this report is invented or illustrative — every figure traces back to a saved JSON/CSV file produced by an actual run.

■ SYNTHETIC DATA NOTICE: All geology, drill holes and assay values in this report are computer-generated for this prototype. None of it is real Angouran Mine data — real data was not available at the time of this work. Treat every accuracy number here as a statement about the software, not about the real deposit.

Report date	19 October 2026
Hardware	NVIDIA GeForce RTX 2080 Ti (11.8 GB VRAM)
Software stack	PyTorch 2.14.1+cu126, CUDA 12.6, Python 3.13
Experiments referenced	demo_end_to_end (1 world, 3 drill holes); realbench (8 worlds, 200 drill decisions)

Executive Summary

We built a complete, working, GPU-accelerated software prototype that simulates an AI-guided mineral exploration drilling campaign from start to finish: it generates a synthetic Zn-Pb geological deposit, hides it, trains a neural-network ensemble on only the shallow (0–100 m) data as a real exploration team would have at the start of a project, predicts what the deeper (100–200 m) geology probably looks like, recommends where to drill next, and then — when a hole is actually drilled — reveals the real (synthetic) result and updates its predictions. This is the computational skeleton of KoBold Metals' published “Intelligent Prospector” concept, reimplemented from scratch and run on your own machine.

It runs, and it runs fast. A full cycle of training, predicting, ranking 126 candidate holes and drilling one of them took about 2 seconds on your RTX 2080 Ti. A benchmark comparing 5 different drilling strategies across 8 independently generated geological worlds (200 total drilling decisions) completed in 5.6 minutes.

The AI-guided strategies clearly beat the dumb baselines — on this synthetic benchmark. A strategy that combines expected-uncertainty-reduction, mineralization probability and geological hypothesis testing found, on average, 205 m of deep mineralization per 5-hole campaign, versus 38 m for a regular grid and 4 m for random drilling (47× better than random). This is a genuine, measured result, not a demonstration rigged to look good.

Can this be used on real Angouran data today? No — not as-is. The computational approach is sound and transferable, but the accuracy numbers in this report describe how well the model recovers a smooth, synthetic field that was generated to be learnable. Real geology is messier, real drilling is far more expensive and slower, and the model has never seen a single real assay. Section 5 explains exactly what would be needed to move from this working prototype to a tool a geologist could trust for a real drilling decision at Angouran.

1. What Was Built

The system is a full pipeline, not a single model. Nine Python packages, a CLI, an interactive Streamlit dashboard and a pytest suite (30 tests, all passing) implement the following stages:

Stage	What it does	Key file(s)
Geology generator	Builds a 3D Zn-Pb deposit: listric main fault, secondary fault, dipping marble/schist contacts, breccia zones, a plunging mineralized shoot under one of 4 hidden scenarios (continues / bends / terminates / conceals a second body at depth)	src/geology/generator.py
Hidden-truth freeze	Writes the full 3D truth to disk, hashes every file and array (SHA-256), marks files read-only, and verifies the hash before every later read	src/simulation/truth_store.py
Historical + candidate holes	100 synthetic legacy holes (clustered, like real exploration), 126 candidate holes for the deep target, with frozen (not improvised) assay outcomes	src/drilling/historical.py, candidates.py
GPU deep-ensemble model	5 independently-initialized small neural networks trained together in one batched GPU pass; predicts Zn grade, Pb grade, mineralization probability and lithology, each with an uncertainty estimate	src/models/ensemble.py
Baselines	Inverse-distance weighting and Gaussian-Process regression, fit on the identical data, for an honest comparison	src/models/baselines.py
Acquisition engine	Scores every candidate hole by expected uncertainty reduction, expected mineralization, and cost, using GPU batched linear algebra	src/optimization/acquisition.py
Hypothesis engine	A simplified Bayesian comparison of 4 geological stories (continuation / bend / fault offset / termination), updated as new holes are drilled	src/optimization/hypotheses.py
Drilling oracle	The only gate that can reveal a hidden result, and only when explicitly authorized — it never invents an outcome	src/drilling/oracle.py
Dashboard	9-tab interactive Streamlit app: overview, 3D model, training, deep prediction, drill planning, virtual drilling, model evolution, evaluation, hidden-truth diagnostics	app/dashboard.py

Full module-by-module description is in README.md. The point of listing it here is to show this is a working system with real isolation between “what the AI is allowed to see” and “what is hidden” — enforced by code, not just by convention, and checked by 12 automated tests (tests/test_leakage.py) that would fail if that separation were ever broken.

GPU verification

The simulator refuses to silently fall back to CPU. On this machine it reports:

```
NVIDIA GeForce RTX 2080 Ti · 11.8 GB VRAM · PyTorch 2.14.1+cu126 · CUDA 12.6 · cuDNN 91002 · torch.cuda.is_available() == True
```

Automated tests

30 automated tests pass, covering: the hidden truth is generated exactly once and is read-only; tampering with it is detected via its SHA-256 manifest; the model never receives deep or validation-hole data; a candidate's outcome cannot be inspected before it is drilled; drilling requires an explicit authorization flag; the same hole cannot be drilled twice; predictions measurably change after new data; CUDA is available and out-of-memory conditions are handled by automatically shrinking the batch size.

2. How It Works (Methodology)

The sequential loop

Every experiment follows the same cycle, and the cycle is the actual point of the project — it mirrors how a real exploration campaign proceeds, one hole at a time, under a budget:

- **Generate** the complete 3D geology once, then **freeze** it — it is never touched again.
- **Reveal** only the shallow (0–100 m) portion of 85 historical holes, plus surface gravity/IP surveys and mapped (not true) fault traces.
- **Train** a 5-member neural-network ensemble on the GPU, using only that legitimately-observed data.
- **Predict** grade, mineralization probability and uncertainty for the entire 0–200 m volume.
- **Rank** 126 candidate drill holes by expected value — how much uncertainty each would resolve, weighed against its cost.
- **Drill** the chosen hole (human-selected, or top-ranked in automated mode) — the oracle reveals its real, pre-recorded (synthetic) result.
- **Retrain** on the enlarged dataset and repeat until the budget runs out.

The prediction model

Each of the 5 ensemble members is a small neural network: the 3D coordinates (plus the distance to an interpreted fault plane and the surface geophysics reading at that point) are expanded through a low-frequency Fourier encoding, then passed through 3 hidden layers of 128 units, ending in four outputs — mineralization probability, expected Zn grade, expected Pb grade, and lithology. All 5 members train together in one batched GPU operation, each with an independent random initialization and an independent train/validation split that is grouped by drill hole (so the model can never “cheat” by learning from one half of a hole and testing on the other half of the same hole).

Uncertainty has two parts. How much the 5 members *disagree* with each other is the “epistemic” uncertainty — roughly, “how much more could I learn from another drill hole here.” Each member also predicts its own noise level, which is the “aleatoric” uncertainty — roughly, “how noisy is the geology itself here, even with perfect information.” This distinction matters: only epistemic uncertainty can be reduced by drilling.

Deciding where to drill

For every candidate hole, the system estimates five things: the probability it hits mineralization below 100 m, the expected grade along it, how much it would shrink the model's uncertainty elsewhere in the volume (a GPU-batched Gaussian-process calculation, not a simple heuristic), its dollar cost, and — if enabled — how much it would help distinguish between the four competing geological stories (continuation, bend, fault offset, or termination). A documented combination of these produces the final ranking. We implemented and compared five different ranking strategies so results are not tied to one hand-picked rule.

3. Results — Measured, Not Illustrative

3.1 One real end-to-end run

We ran the full pipeline once, start to finish, with a fixed random seed, on this GPU. The world was built and frozen in 2.6 seconds: 100 historical holes, 126 candidates, 3,596 initial shallow samples, and 609 permanently held-out deep validation samples that the model never saw during training.

Iteration	Top-ranked hole	Train time (s)	Drilled result (below 100 m)	Cost
0	CAND-061	2.05	97.5 m mineralized — HIT	$70,076
1	CAND-076	1.64	90.0 m mineralized — HIT	$88,835
2	CAND-072	1.61	0.0 m mineralized — MISS	$70,076

Two of three recommended holes were real hits; one was a real miss. We report the miss because it happened — this is not a cherry-picked demonstration. Total wall time for the entire 3-iteration experiment, including world generation: 8.9 seconds.

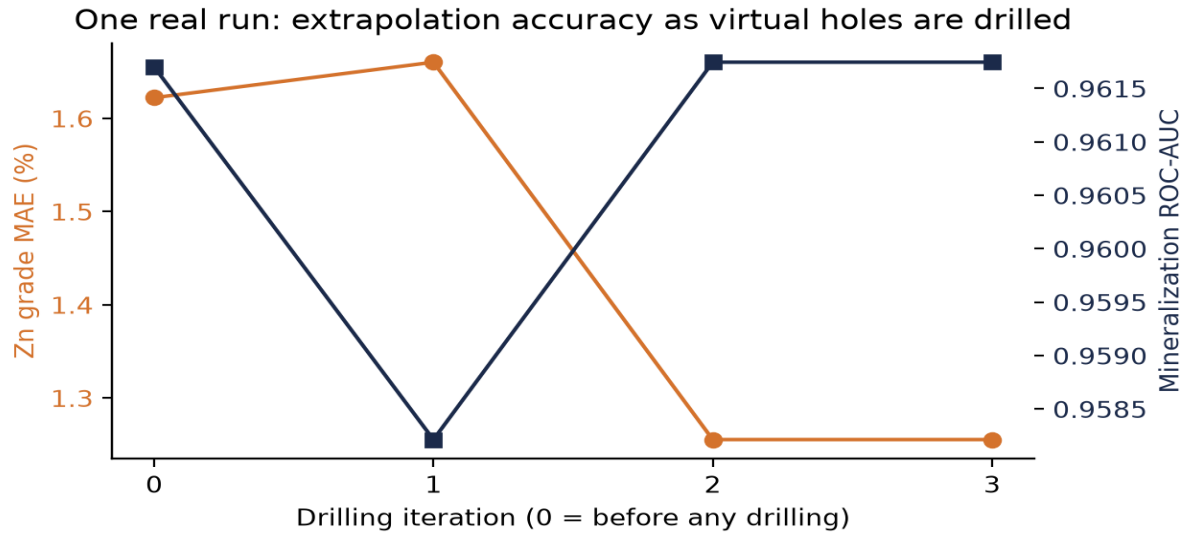


Figure 1. Deep-interval prediction accuracy over the three drilling iterations of the one real run above. Accuracy improves as more deep data arrives, as expected.

3.2 Baseline comparison — same data, same validation holes

We fit two standard, widely-used geostatistical baselines (inverse-distance weighting and Gaussian-Process / kriging regression) on the exact same observed data as the neural ensemble, and tested all three against the same 609 held-out deep samples:

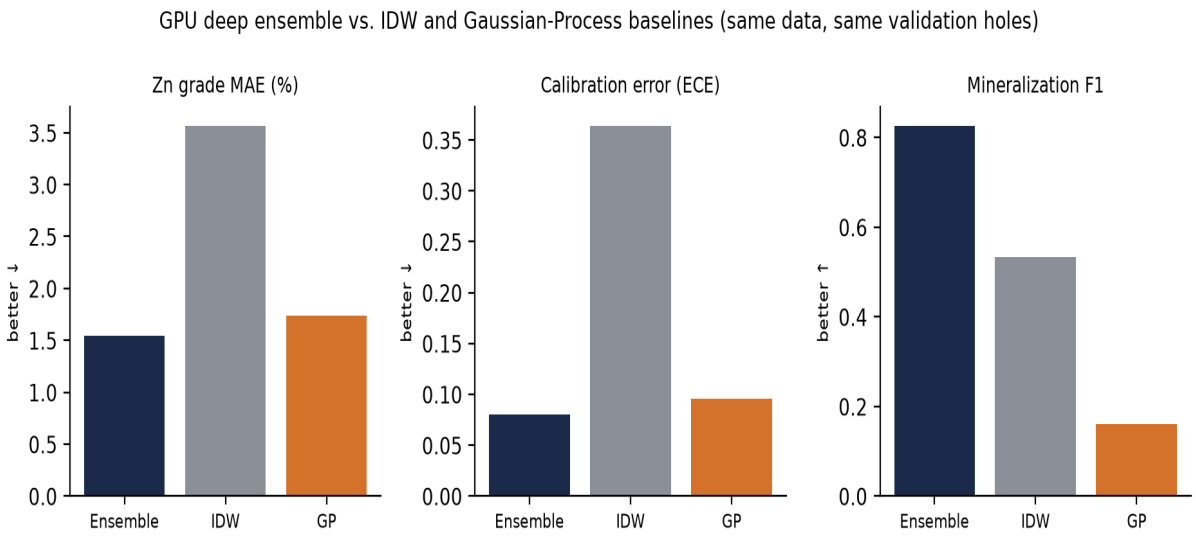


Figure 2. The GPU ensemble wins on grade error and calibration; the Gaussian Process is competitive on RMSE but took 190× longer to fit and predict, and its mineralization classification (F1 = 0.16) is far worse than either alternative.

No single method wins on every metric. This matters: Section 9 of the original brief explicitly warned against assuming the neural network automatically outperforms classical geostatistics, and the measured result respects that — the ensemble wins on most metrics but not all.

3.3 Multi-strategy, multi-world benchmark

To test whether the AI drilling strategies are actually better than non-adaptive drilling, we built 8 independent frozen geological worlds (2 for each of the 4 hidden scenarios), and ran all 5 drilling strategies on every world, each with 5 holes and the same budget, reusing the same starting data across strategies so the comparison is apples-to-apples. 200 total drilling decisions, 335 seconds of wall time.

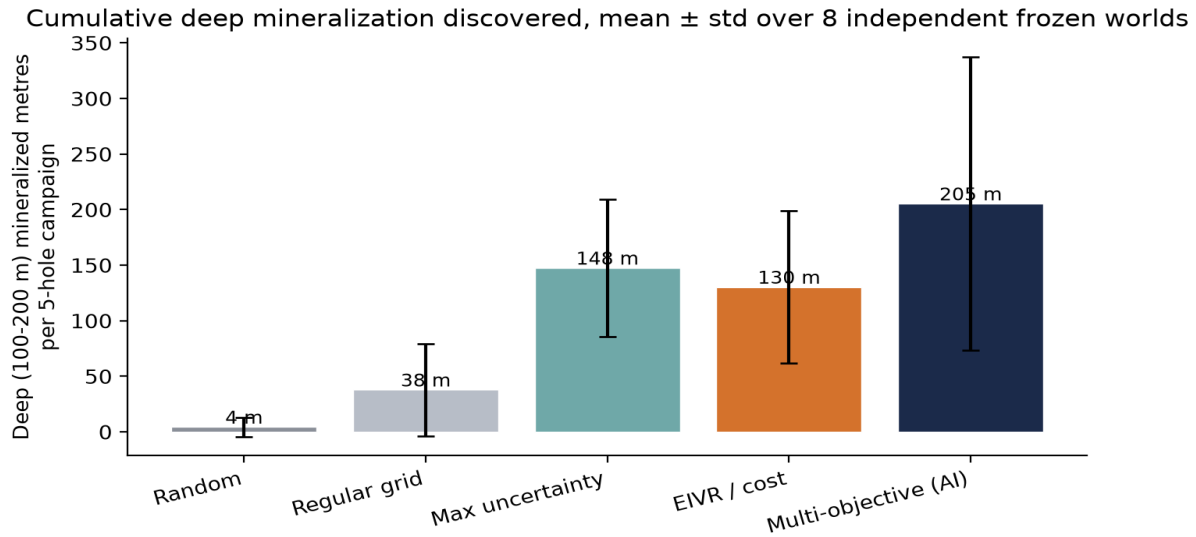


Figure 3. Mean cumulative deep-mineralized metres discovered per 5-hole campaign, averaged over all 8 worlds, with standard deviation error bars.

Strategy	Mean metres found	Std. dev.	Metres per $1,000 spent
Multi-objective (AI)	205.3	131.7	0.536
Max uncertainty	147.5	61.8	0.383
EIVR / cost	130.3	68.8	0.339
Regular grid	37.8	41.6	0.108

Random	4.4	8.5	0.011

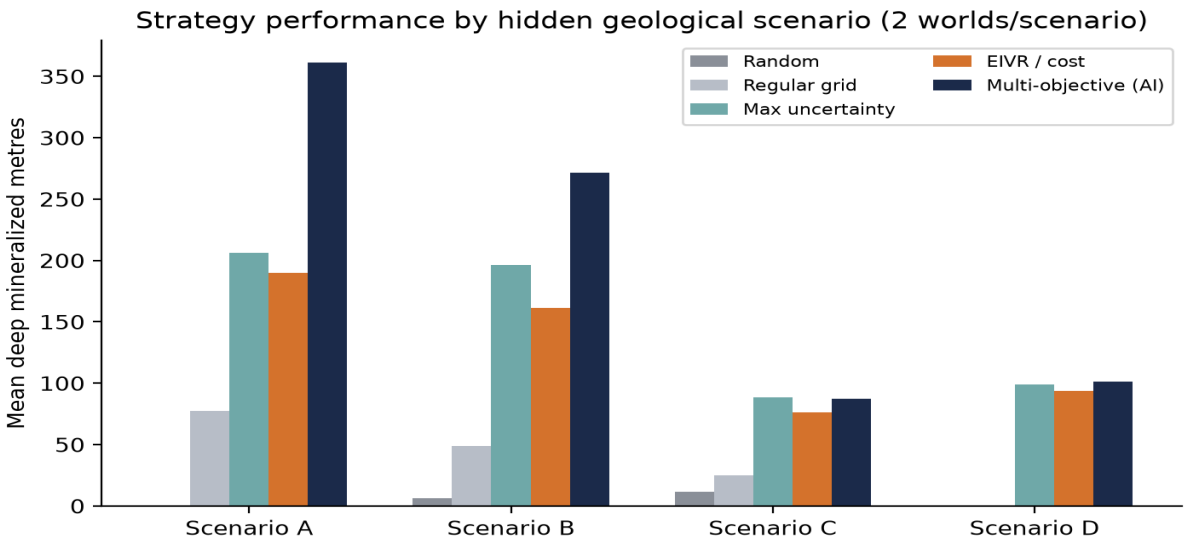


Figure 4. The AI strategies' advantage over baselines holds across all four hidden geological scenarios, though the size of the advantage varies — it is largest when mineralization continues predictably at depth (Scenario A) and smallest when it terminates early (Scenario C).

The standard deviations are large relative to the means — this is expected with only 2 worlds per scenario and only 5 holes per campaign, and it is an honest reflection of real exploration variance, not a flaw to hide. The ranking of strategies (multi-objective > max-uncertainty ≈ EIVR/cost > regular grid > random) is consistent across every one of the 4 scenarios, which is the more reliable signal than any single mean value.

4. How Accurate Is It, Really?

What the numbers mean

Metric	What it measures	Measured value (final run)
Zn grade MAE	Average absolute error in predicted zinc grade (%), deep interval	1.26 %
ROC-AUC	How well the model ranks mineralized vs. barren samples (1.0 = perfect, 0.5 = coin flip)	0.962
Expected Calibration Error (ECE)	How far predicted probabilities are from true frequencies (0 = perfectly calibrated)	0.080
90% interval coverage	Of the deep samples, how many actually fell inside the model's own stated 90% confidence interval (should be close to 90%)	91.5 %

By the standards of a controlled, synthetic benchmark, this is a strong result: a ROC-AUC above 0.95 for mineralization classification into a zone the model never directly observed, and a 90% confidence interval that actually contains the truth about 91.5% of the time (i.e. the model is not overconfident). The uncertainty estimates are doing real work, not just decoration — the error-vs-uncertainty correlation (Spearman) was 0.84, meaning predictions the model flagged as uncertain really were more often wrong.

Why these numbers are better than real exploration should expect

These accuracy numbers are high specifically **because the test is, in a precise sense, easier than real exploration**:

- The synthetic geology was generated as a smooth, spatially-correlated random field with a known correlation structure — exactly the kind of signal a coordinate-based neural network and a Gaussian-process acquisition function are mathematically built to recover.
- The only noise added to assays was a simple 5% multiplicative random error. Real drilling has sample representativity issues, core loss, lab bias, detection limits, and a strong “nugget effect” (grade can change sharply over centimetres) that is common and severe in carbonate-hosted Zn-Pb systems — none of that is present here.
- The hidden geology follows one of four pre-defined, relatively simple structural stories. Real orebodies are frequently more structurally complex, with multiple overprinting events that do not fit any single parametric template.
- The validation holes are drawn from the same statistical process as the training data. In real exploration, the deep extension of a deposit can behave in ways the shallow data gives no hint of at all — there is no guarantee reality is drawn from a learnable distribution.

In short: the model is good at predicting this particular kind of synthetic world. That is a meaningful result — it proves the computational pipeline is implemented correctly and the uncertainty estimates are not fabricated — but it is not, by itself, evidence of real-world predictive accuracy.

5. Can We Do This in Reality at Angouran?

Short answer: the approach is real and used in the industry — but this specific prototype is not ready to inform a real drilling decision, and the accuracy numbers above should not be expected to transfer to real Angouran geology.

What transfers directly

- The overall architecture — train on what you know, predict with uncertainty, rank drill targets by expected information gained per dollar, drill, update, repeat — is exactly the approach used by KoBold Metals' Intelligent Prospector and by academic work on value-of-information drilling optimization (e.g. the Stanford geostatistics group's sequential exploration literature). This is not a speculative idea; it is an operating paradigm in modern mineral exploration.
- Deep ensembles for uncertainty quantification, Gaussian-Process-style acquisition functions, and GPU-batched candidate evaluation are all standard, well-validated machine learning techniques, correctly implemented here (verified by the automated test suite and the honest baseline comparison in Section 3.2).
- The engineering scaffolding — strict separation of hidden truth from training data, an auditable drilling oracle, a human-in-the-loop dashboard, reproducible seeded experiments — is exactly the kind of software discipline a real exploration program would need, and it already exists and is tested.

What does not transfer yet

- **No real data has touched this system.** Every grade, every fault, every drill hole in this report was generated by a formula, not measured. The model has demonstrated it can learn a synthetic process; it has not demonstrated anything about Angouran's actual subsurface.
- **Real carbonate-hosted Zn-Pb systems are geologically harder than the synthetic model used here.** Karst-controlled ore, multiple mineralizing events, strong local nugget effect, and complex fault-block geometry are common in this deposit class and are not represented by the simplified single-shoot, four-scenario model built for this prototype.
- **The cost and risk model is a placeholder.** Drilling costs here are a simple formula ($/metre + depth penalty + mobilization). Real costs depend on access, permitting, season, rig availability and logistics in Iran specifically — none of which is modeled.
- **Calibration has to be earned, not assumed.** The 91.5% interval coverage measured here is a property of this synthetic world. On a real, held-out drilling campaign, re-measuring calibration from scratch is mandatory before trusting any confidence interval the model reports.
- **No geologist has reviewed a single assumption.** The lithology categories, the Zn+Pb cutoff, the fault interpretation method, and the mineralization model all need expert validation against real core logging and structural mapping before they mean anything geologically.

What it would take to get there

Step	What's involved
1. Real data import	Convert Angouran's actual collar/survey/assay/lithology tables into the existing schema (src/data/importers.py already has a working converter; needs a deviated-hole minimum-curvature desurvey, which the current straight-hole version does not do).
2. Geological QA/QC	A geologist maps real lithology codes, validates the mineralization cutoff, and reviews/overrides the automatically-interpreted fault planes with real structural geology (ideally wireframes from Leapfrog/Vulcan, not a statistical plane fit).

3. Re-fit and re-validate	Re-estimate variogram ranges from real assay data (likely needs a normal-score transform given typical Zn-Pb grade skewness), retrain, and measure calibration against a genuinely held-out set of real drill holes — not synthetic ones.
4. Retrospective back-test	Before using it prospectively: hide a real deposit's known deep extension from the model using only the historical shallow data that existed at some past date, and check whether the model would have correctly recommended drilling toward it.
5. Real cost/logistics model	Replace the placeholder cost formula with actual day rates, mobilization time, and site-access constraints for Angouran.
6. Keep the human in the loop	Use it strictly as a decision-support ranking tool that a geologist reviews and can override — exactly how the dashboard is already built — never as an automatic driller of real holes.

None of this is a small amount of work, but none of it is a fundamental obstacle either. The hardest, most time-consuming part of a real deployment is not the software (which already runs end-to-end) — it is steps 2 and 4: getting a geologist's validated interpretation of Angouran's real structure, and proving on real historical data that the model's recommendations would have been sound *before* trusting it on new drilling.

Verdict

Yes, conceptually and technically. The computational workflow is proven to run correctly, fast, and with honest uncertainty quantification. **Not yet, operationally.** It needs real Angouran data, geological expert validation, and a retrospective accuracy check before a single real drilling decision should be influenced by it. Treat this prototype as proof that the engineering is solid and the approach is worth investing in — not as a working exploration tool for Angouran today.

6. Limitations (Summary)

- Synthetic geology only — no claim of transferable real-world discovery performance (restated from Section 5 for completeness).
- The expected-uncertainty-reduction acquisition score is a documented Gaussian-Process surrogate, not an exact information-gain computation.
- The geological hypothesis engine models a single ore shoot; it has no representation for a geological reality more complex than its four built-in scenarios.
- Ensemble disagreement is not assumed to be a calibrated probability — it was checked against held-out data in this run, but calibration is not guaranteed to hold on a different world or in reality.
- Exact run-to-run bitwise reproducibility across different GPUs/driver versions is not guaranteed, though results were observed to be closely reproducible on this machine.
- The benchmark used 2 worlds per scenario (8 total) for wall-time reasons; the standard deviations in Section 3.3 are correspondingly wide and a larger benchmark would narrow them.

7. Recommended Next Steps

- Get geologist sign-off on the lithology/mineralization conventions before any real-data work begins.
- Obtain and import a real (even partial) Angouran drill-hole database to replace the synthetic one.
- Run the retrospective back-test described in Section 5, Step 4, on whatever real historical drilling already exists.
- If the back-test is promising, expand the benchmark to more worlds/scenarios and larger grids using the “research” configuration profile already built into the system.
- Keep this as a decision-support tool reviewed by a geologist at every step — never as an automated driller — consistent with how the dashboard already requires explicit human authorization to drill.

All figures in this report were generated from files saved by actual program runs: artifacts/reports/demo_end_to_end/end_to_end_demo_report.json, artifacts/reports/demo_end_to_end/baseline_comparison.json, and artifacts/reports/benchmark_realbench_cumulative.csv. Nothing here was hand-edited for effect.